Did Expert Iteration Create It?

A Pretraining Knockout and Compute-Matched Budgets
in a From-Scratch Theorem Prover

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Draft, October 13, 2026. Not for circulation.

Abstract

Whether reinforcement learning (RL) against a verifier creates capabilities or raises the probability of behaviour the base already had is contested; large-model studies can rarely change what the base saw. We study it in a 9.56M-parameter transformer prover trained from scratch on 155,000 synthetic propositional-logic proofs and judged only by Lean 4. We remove one inference composition from pretraining, the classical reductio σ (assume $\neg G$, refute it, eliminate the double negation), keep both of its rules, refill the data to the same size, and run 16 rounds of expert iteration (EI), a rejection-sampling form of RL. No knockout seed acquired a classical theorem family (half its held-out members proved in 256 attempts): 0/6 against 6/6 full-exposure seeds, a pre-registered contrast ($p = 0.00024$) largely present before RL. During RL, full-exposure seeds gained 8–69 held-out classical theorems and knockout seeds 0–2; 5/6 against 0/6 acquired a new family ($p = 0.0032$; not pre-registered). Restoring σ in 1% of pretraining restored acquisition (3/3 seeds). The knockout bases were not σ-free: five of six proved held-out theorems with σ 1–7 times in 1.1–1.5 million attempts. EI trained on their σ proofs of easier theorems from round 1 but did not carry the step to the theorems that need it. Measurement matters too: in three earlier full-exposure runs, 51–60 theorems per run look RL-created at equal sample counts, and 14–21 remain when the base gets RL's compute as attempts. In this organism, EI amplified a composition pretraining had exposed but did not acquire it within 16 rounds from bases pretrained without it.

1 Introduction

Reinforcement learning against a verifier is now the standard way to improve reasoning models. Whether it teaches them anything new is contested. Yue et al. [56] find that base models overtake their RL-trained descendants in pass@k as k grows. Karan and Du [21] approach RL's gain on a mathematics benchmark by sampling the base from a sharpened distribution, with no training. Other work reports RL solving problems the base never solved [1, 27]. Part of the disagreement is about budgets. A base whose per-attempt success probability on a problem is 10^{-6} will almost never solve it in 1,024 attempts, yet the capability is present in a sense that matters for evaluation and safety. A capability is defined only relative to a sampling budget [5, 26].

A claim that RL *created* a capability therefore needs three things:

- a budget at which the base is judged unable, tied to what RL spent;
- controls that receive RL's training without RL's proofs;
- ideally, a base that never had the capability [9, 18].

Pretraining interventions before RL exist. Zhang et al. [58] vary a context's share of pretraining (0%, 0.1%, 1%) in a 100M model and find no RL transfer below 1%. Abdulsalam et al. [1] exclude

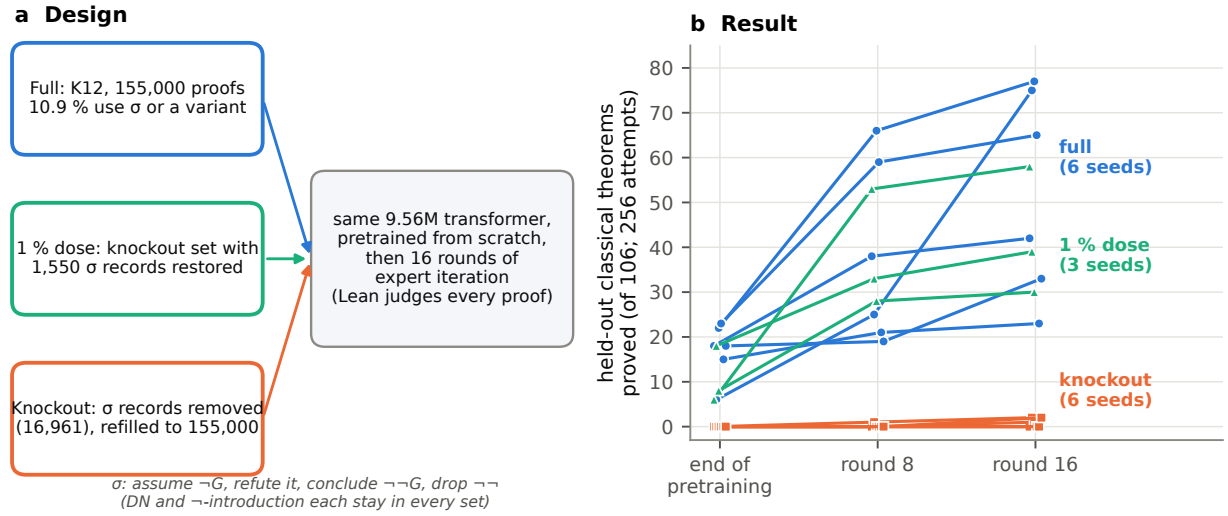


Figure 1: Design and result. (a) Three pretraining sets that differ only in the classical reductio σ. (b) Held-out classical theorems proved (of 106, with 256 attempts each) at the end of pretraining and after 8 and 16 rounds of expert iteration, one line per seed. No knockout seed acquires a classical family in 16 rounds.

shortcuts from pretraining and see GRPO build them. Tsilivis et al. [43] set the share of long demonstrations exactly. We add a narrower intervention, with the base's own rate measured. We remove a single inference composition from pretraining, keep both of its parts, refill the data to the same size, and measure the base's rate for the composition at about a million attempts.

We do this in a model organism small enough to allow it. It is a 9.56M-parameter transformer prover for propositional logic, pretrained from scratch on 155,000 synthetic natural-deduction proofs. It acts through a proof-state interface, one inference step at a time, and Lean 4 alone decides whether a proof counts. The composition we remove is the classical reductio σ: to prove a positive goal G, assume $\neg G$, derive a contradiction, conclude $\neg\neg G$ by negation introduction, and eliminate the double negation (`DN`, Lean's `Classical.byContradiction`). Both rules stay in the data, used separately. We track six classical theorem families, such as excluded middle and Peirce's law, whose *strict* members can only be proved by eliminating a double negation the proof itself built. Their 106 held-out strict theorems are never trained on. We then run expert iteration (EI) from each base: sample, keep what Lean accepts, fine-tune, for 16 rounds (Fig. 1).

Two objections, and the null model. The first objection is that a model cannot learn what it never saw. But the knockout bases saw both rules, and they still write σ: five of six proved held-out strict theorems with it at large sample sizes. The second objection is stronger. At the bases' measured rates, EI's 3,936 attempts per round on the 123 strict training theorems should find at most 0.02 σ proofs per round. A first success would then be expected only after 47 rounds or more. We take this waiting-time arithmetic as the null model and ask what happens beyond it.

Results.

- **No knockout seed acquired a classical family in 16 rounds** (0/6, which bounds the per-seed rate below 0.39), against 6/6 full-exposure seeds. That pre-registered contrast is largely decided before RL: one family is already acquired at the end of pretraining on 5/6 full seeds. The contrasts specific to RL were not pre-registered. During RL, full seeds gained 8–69 held-

out strict theorems and knockout seeds 0–2 (exact Mann–Whitney $p = 0.0011$). Five of six full seeds against none acquired a family during RL ($p = 0.0032$; on the three seeds that shared the knockout's hardware, 2/3 vs 0/6, $p = 0.034$).
- **One percent is enough.** Restoring σ in 1% of pretraining records restored acquisition on 3/3 seeds.
- **Beyond the null.** The knockout ladders wrote σ from round 1 on theorems that supply the double negation, and trained on 844–4,428 such proofs. Three of six reached a strict training theorem sooner than their bases' rates predict: 0.55 seeds expected by round 16, 3 observed, $p = 0.006$ under the null. None carried the step to a family. The ladder that went furthest, continued to 24 rounds, proved 5 of the 106 held-out theorems, more often each round, and no more of them.
- **Measurement.** In three earlier full-exposure runs, 51–60 theorems per run look RL-created at equal sample counts. Only 14–21 remain when the base gets RL's compute as attempts, under every budget allocation we tried.

Contributions.

1. **A composition knockout in pretraining** (§3), with a dose arm and diagnostics that separate "absent" from "rare": base rates at about 10^6 attempts, certificates, a waiting-time null, and likelihoods of known proofs along every RL run. It was pre-registered, with six seeds per main arm, and an analyst who did not write the experiment's code recounted it independently.
2. **A compute-matched measurement protocol, applied end to end** (§2, §4). It gives the base a budget of attempts matched to RL's compute under several defensible allocations, so that the "created" set becomes a curve in the budget. It brackets each theorem's success rate between sampling bounds and a likelihood estimate, certifies unreachability, and nets out RL-free training. It integrates budget-relative definitions from prior work [8, 26, 45].
3. **An open organism**: a from-scratch prover with a sound judge, likelihoods of arbitrary proofs at every checkpoint, and pretraining that can be edited one composition at a time. Code, data, checkpoints and per-arm compute are public.

What transfers. The protocol and the waiting-time logic transfer to larger models; the numbers do not. The 1% threshold, the 10^{-6} rates and the dynamics of EI belong to this organism, and they depend on how many training problems need the composition. Two lessons follow, with that caveat. Failure at modest k does not show absence: our knockout bases fail at $k = 256$, yet succeed at pooled rates near 10^{-6} per attempt, and at up to about 5×10^{-4} on a few theorems. And removing a composition from pretraining kept this RL from acquiring it within 16 rounds, but 1% of the data restored it. Section 6 discusses what other algorithms and injected demonstrations did to similar patterns in earlier, smaller organisms.

2 The organism and the experiment

Logic, data and judge. Theorems are sequents $\Gamma \vdash G$ of propositional logic over at most four atoms, with $\wedge, \vee, \to, \neg, \bot$. Proofs are Fitch-style natural deduction: premises, assumptions that open boxes, the introduction and elimination rules of each connective, $\bot$-elimination and double-negation elimination (`DN`). The pretraining set, which we call K12, holds 155,000 synthetic proofs of 2–12 lines from a generator, with a flat length distribution. A proof counts if and only if Lean 4 core accepts it. A syntactic pre-filter may reject malformed text before Lean, but never accepts anything on its own (§3.8).

Model and interface. Every model is the same decoder-only transformer: 6 layers, width 384, ALiBi and NoPE attention heads, 9,560,832 parameters. It is pretrained from scratch for 1,200 s on one NVIDIA A40, about 24,000 steps or 3.7 passes over the (state, action) pairs of K12. It acts in a *proof-state environment*. It sees the focused goal's Lean tactic state and writes one step, such as `have n5 : ¬P := n1 n4`. The environment applies the step, opens and closes the boxes of →- and ¬-introduction and the branches of ∨-elimination, and ends the attempt at the first step it cannot apply. It also numbers the lines from one of 33 random starting offsets, so every proof has 33 spellings. The finished proof is rendered in Lean and judged.

Expert iteration. Each round, the current model samples $k = 32$ attempts at temperature $T = 0.8$ on each of 4,495 training theorems. Accepted proofs accumulate across rounds, up to four per theorem. The model is then fine-tuned for 600 steps, from where it left off, on those proofs (each repeated four times) mixed with 20,000 records replayed from its own pretraining set. The training budget is the same for every theorem. This is rejection-sampling fine-tuning in the lineage of expert iteration [3, 32], STaR [57] and ReST^{EM} [40]. We call one 16-round run from one pretrained seed a *ladder*, and the model after N rounds rN; r0 is the pretrained base. Each round also samples $k = 32$ attempts on every held-out family member, which is never trained on.

Evaluation. An evaluation draws $k = 256$ independent attempts per theorem at $T = 0.8$, with at most 96 steps and 512 tokens per step. A theorem is *proved* if any attempt is accepted. Two sample seeds give two independent draws of the same checkpoint. A *guided* evaluation uses the same budget, but a step that a per-step logical checker rejects is redrawn from the same state without replacement, up to 10 times per attempt. The checker never rejected a step Lean accepts in 314,541 tested (state, step) pairs. General ability is measured on 5,000 held-out generator theorems decoded greedily, and on 322 evaluation theorems (72 textbook problems plus 250 held-out generator theorems; §4).

The composition and the theorem families. Table 1 defines the terms used throughout. In K12, σ occurs in 16,703 records and σ^+ in 16,961 (10.94%); the lemma route occurs in none. We track six classical generator families: converse contraposition ($\neg B \to \neg A \vdash A \to B$), negated conditional ($\neg(A \to B) \vdash A \wedge \neg B$), Peirce's law, the Peirce sequent ($((A \to B) \to A \vdash A$), excluded middle, and De Morgan ($\neg(A \wedge B) \vdash \neg A \vee \neg B$). Each has 40 training members and 40 held-out members, substitution instances of the family's schema. The strict members number 123 in training and 106 held out: 105 family members plus one generated theorem, `la_transfer_916` (Table 2). A model *acquires* a family if it proves at least half of the family's strict held-out members in one evaluation. Figure 2 shows a σ proof written by a knockout base.

Arms. The arms share the network, the pretraining recipe and the EI protocol (Table 3).

- **knockout** pretrains on K12$^-$. This is K12 minus exactly its 16,961 σ^+ records, refilled with 16,961 records drawn length by length from K12's own unused source pools. Every refill record passes K12's filters and contains no σ^+ and no lemma route. Under two renaming-invariant keys, the refill shares no theorem with K12 or with any evaluation file. Lean accepts all 16,961 refill records.
- **dose** restores 1,550 removed σ records in place of refill records, at K12's length mix: 1.00% of the records.
- **full** pretrains on K12. Seeds s0–s2 come from earlier runs. They differ from the new seeds s3–s5 in three ways. Rounds 1–8 ran on RTX A6000 cards at sampling batch 2,048, whereas the new

Table 1: The terms that separate "absent" from "rare". A *derived* line is one produced by a rule, not a premise or an assumption.

term	definition
σ (classical reductio)	`DN` applied to a $\neg\neg G$ that the proof built by $\neg$-introduction (assume $\neg G$, derive $\bot$). In Lean: `Classical.byContradiction` on that refutation.
σ^+	`DN` applied to any derived line (σ plus routes through $\to$-introduction, $\vee$-elimination, …).
lemma route	prove $\neg\neg G \to G$ inside an implication box (`DN` on the box's assumption), then apply it to a $\neg\neg G$ built by $\neg$-introduction. Classical, but not σ^+.
free positive reductio	a σ^+ step whose conclusion G is not a negation and whose $\neg\neg G$ does not occur in the theorem. It is the genuinely classical use; using σ on a theorem that supplies $\neg\neg G$ is not.
classical-only	the theorem has no intuitionistic proof (decided by the complete G4ip procedure).
strict member	a classical-only family member with no $\neg\neg$ subformula in its sequent. Every proof of it must eliminate a double negation that the proof built: by a free positive reductio or by the lemma route.

Table 2: Strict members per family. The held-out total adds one generated theorem, `la_transfer_916`.

	conv. contrap.	neg. cond.	Peirce	Peirce seq.	excl. middle	De Morgan	total
training (of 40 each)	27	16	15	18	38	9	123
held-out (of 40 each)	18	15	12	19	37	4	105 + 1

seeds ran on A40s at 1,024. The ladders were resumed at rounds 9 and 13, which restarted the replay sampler. And each round's evaluation phase sampled more theorems. Changing the batch changes which samples are drawn, not how they are judged.

Because each ladder replays its own pretraining data, exposure differs during RL as well. Full ladders replay about 2,200 σ^+ records per round, dose ladders about 200, and knockout ladders none. "Exposure" below therefore means pretraining together with the replay drawn from it. Arms with the same seed number share initial weights and sampling seeds, not data.

Budgets, reach and certificates. We write $p_M(t)$ for the probability that one attempt of model M on theorem t is accepted, and $\hat{p} = c/n$ for its estimate from c successes in n attempts. For a budget of K attempts per theorem:

$$\begin{aligned} t \text{ within reach of the base at } K &\iff \hat{p} \geq 1/K, \\ t \text{ not reached at } K &\iff c = 0 \text{ with } n \geq K, \\ t \text{ certified out of reach at } K &\iff \mathrm{UB}_{95}(c, n) < 0.05/K, \end{aligned}$$

where UB_{95} is the one-sided 95% Clopper–Pearson bound; with $c = 0$, certification needs $n \approx 60K$. "Within reach" from a single success in fewer than K attempts is weak evidence, and we flag it where it occurs. A *pooled* certificate over a set of theorems bounds their mean per-attempt rate, not any one of them. K is matched to RL's compute as $K = C_{\mathrm{RL}}/(N_{\mathrm{eval}}\, c_{\mathrm{base}})$. Here C_{RL} is the GPU time of the ladder, or the part of it an allocation charges (§4), and c_{base} is the base's measured cost per attempt. $N_{\mathrm{eval}} = 322$ is the number of evaluation theorems the budget is spread over. N_{eval} is a choice: spreading the same compute over the 1,054 theorems of our largest evaluation would make every K 3.3 times smaller. The *compute-matched set* $\mathcal{C}(K)$ holds the theorems that RL proves and

```
theorem t (P Q R S : Prop) (h1 : ((¬S) → (¬P))) (h2 : P) : (R → S) := by
  have n1 : ((¬S) → (¬P)) := h1
  have n2 : P := h2
  have n9 : (R → S) := (fun (n3 : R) => by
    have n7 : (¬(¬S)) := (fun (n4 : (¬S)) => by
      have n5 : (¬P) := n1 n4
      have n6 : False := n5 n2
      exact (n6 : False))
    have n8 : S := Classical.byContradiction (fun hh => n7 hh)
    exact n8)
  exact n9
```

Figure 2: A free positive reductio written by a knockout base before any RL: $\neg S \to \neg P,\ P \vdash R \to S$ (`la_transfer_916`), one of the rare successes of knockout seed 1's base in large-k sampling. Assume $\neg S$ and refute it (`n7`), then eliminate the double negation (`n8`). Lean accepts it and rejects the same proof of the negated goal. Appendix E gives two more.

Table 3: Arms. Every model: 9,560,832-parameter transformer, proof-state interface, pretrained from scratch, then 16 rounds of expert iteration ($k = 32$, $T = 0.8$, 600 fine-tuning steps per round, 4,495 training theorems). Lean 4 alone judges.

arm	pretraining set (155,000 records each)	seeds	σ^+ share
FULL	K12	6 (s0–s5)	10.94%
KNOCKOUT	K12$^-$: K12 − 16,961 σ^+ records + 16,961 refill	6 (s0–s5)	0
DOSE	K12$^{1\%}$: K12$^-$ with 1,550 σ records restored	3 (s0–s2)	1.00%

its base cannot reach within K. $\Delta_{\text{cov}}(K)$ is the number RL proves minus the number the base reaches.

The pre-registered decision rule. Per family, at r16:

- CREATED if at least 2 of 6 knockout seeds acquire it and the knockout base is certified out of reach on it;
- NEEDS EXPOSURE if at most one knockout seed acquires it and more full seeds do (one-sided Boschloo test, $p \leq 0.05$);
- NEITHER if no seed acquires it;
- UNDETERMINED otherwise.

The primary test compares, per seed, the indicator "at least one family acquired at r16" between the full and knockout arms. The pre-registration fixed the certificate pool as the 106 strict held-out theorems, at each knockout seed's training-phase budget $K_{\text{train}}(16)$. As sampled, about 10^4 attempts per theorem, no single family could be certified at that budget, so CREATED could not be returned. Section 3 reports what the remaining verdicts can and cannot say.

3 Results: the σ knockout

All numbers in this section are on the 9.56M-parameter models of Table 3, judged by Lean alone, from the evaluation with sample seed x0 unless stated. An analyst who had not written the experiment's code recounted every count in this section from the raw files, with separate code, and re-checked the proofs in Lean. Numbers computed for this paper are marked (*this paper*); they are recomputed from those recounted files.

Table 4: Strict held-out theorems proved (of 106) per seed at r0 (end of pretraining), r8 and r16, and the families acquired. x0 and x1 are two sample seeds; "guided" is the guided evaluation. Families: cc = converse contraposition, nc = negated conditional, pe = Peirce, ps = Peirce sequent, em = excluded middle; families acquired already at r0 are underlined. † = seeds from earlier runs (other GPU class and batch in rounds 1–8). The dose arm's other evaluations are in Appendix I.

model	r0	r8	r16 x0	r16 x1	guided	acquired at r16 (x0)	gain r0→r16
FULL s0†	18	38	42	43	44	cc, nc	24
FULL s1†	6	25	75	73	75	cc, em, pe	69
FULL s2†	22	66	77	77	80	cc, nc, pe, ps	55
FULL s3	23	59	65	64	64	cc, nc, pe, ps	42
FULL s4	15	21	23	22	24	cc	8
FULL s5	18	19	33	34	33	cc, ps	15
KNOCKOUT s0	0	0	0	0	0	–	0
KNOCKOUT s1	0	1	2	2	3	– (r20: 5; r24: 5)	2
KNOCKOUT s2	0	0	1	1	0	–	1
KNOCKOUT s3	0	0	0	0	0	–	0
KNOCKOUT s4	0	0	0	0	0	–	0
KNOCKOUT s5	0	0	2	1	2	–	2
DOSE s0	6	53	58			cc, nc, pe, ps	52
DOSE s1	18	33	39			cc, nc	21
DOSE s2	8	28	30			cc, nc	22

3.1 No knockout seed acquired a family in 16 rounds

No knockout seed acquired any classical family by r16; every full seed had at least one (Table 4, Fig. 3). This is the pre-registered primary indicator, 6/6 against 0/6, one-sided Boschloo $p = 0.00024$. The same holds in the second sample seed, in the guided evaluation, and with the looser held-out sets first pre-registered (178 theorems; they were replaced by the strict sets in a dated amendment after round 1 of the first knockout ladder, because that base proved 136 of 155 generated "classical" targets using a $\neg\neg$ the sequent supplied). The 0/6 bounds the per-seed rate at which this EI acquires a family from a σ-free base below 0.39 (one-sided 95%).

The indicator is a weak statement about RL, however. Converse contraposition is acquired *before* RL on 5/6 full bases (6/6 in the guided evaluation). Applied to the bases, the indicator already gives 5/6 against 0/6 ($p = 0.0032$), or 6/6 against 0/6 in the guided evaluation ($p = 0.00024$). Per family, the pre-registered rule gives NEEDS EXPOSURE only for converse contraposition (6 vs 0 seeds; Holm-corrected over six families, 0.0015). It gives UNDETERMINED for negated conditional, Peirce and the Peirce sequent (3 vs 0 each; $p = 0.034$, Holm 0.17) and for excluded middle (1 vs 0), and NEITHER for De Morgan, whose four strict members no model proves.

The comparisons specific to RL were not pre-registered. During RL, the full seeds gained 24, 69, 55, 42, 8 and 15 strict held-out theorems and the knockout seeds 0, 2, 1, 0, 0 and 2: the arms do not overlap (exact one-sided Mann–Whitney $U = 36$, $p = 0.0011$). Counting only families acquired between r0 and r16, it is 5/6 against 0/6 ($p = 0.0032$). The old full seeds ran rounds 1–8 on other hardware. Restricted to the three new full seeds, it is 2/3 against 0/6 ($p = 0.034$). On the training side, the knockout ladders proved none of the 123 strict training theorems in round 1 and only 0, 5, 1, 0, 0 and 2 of them by round 16. The full ladders went from 21, 1, 27, 17, 15 and 20 in round 1 to 49, 80, 86, 78, 33 and 36 (Fig. 4a). No knockout ladder ever had proofs for half of any family's strict training theorems.

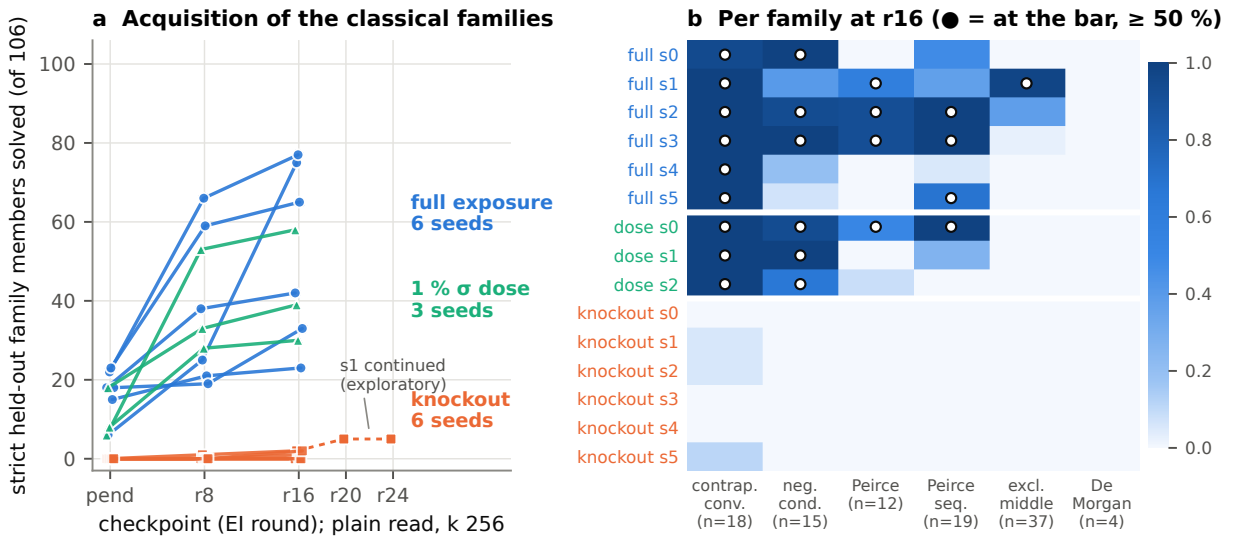


Figure 3: Per seed and per family. (a) Strict held-out theorems proved at r0, r8 and r16; knockout seed 1 was continued to r24 (dashed; exploratory). (b) Share of each family's strict held-out members proved at r16; a dot marks an acquired family ($\geq 50\%$).

3.2 One percent is enough

The dose bases proved 6, 18 and 8 strict held-out theorems at r0, and 58, 39 and 30 at r16. All three acquired families during RL; dose s1 had already acquired converse contraposition at r0. That is 3/3 against the knockout's 0/6 (Boschloo $p = 0.0033$; this arm was added by a dated amendment before it ran, and is exploratory). At r16 the dose seeds lie inside the full arm's range (two-sided Mann–Whitney $p = 0.55$ with three seeds; *this paper*). In round 1 the dose bases wrote 223, 183 and 241 σ proofs, and the free positive reductio from round 1 on. The new full seeds wrote 635–885 σ proofs and the knockout seeds 6–13. The dose changes the data around σ far less than the knockout removed. In $\mathrm{K12}^{1\%}$, `DN` appears in 4.24% of records and ¬-introduction in 6.12%, against 3.29% and 5.19% in $\mathrm{K12}^{-}$ and 13.73% and 15.21% in K12. Classical-only theorems are 1.56% of a 20,000-record sample, against 1.35% and 4.17% (*this paper*, from the dose set's file). A second sample seed and a guided evaluation of the dose arm, run for this paper, agree. The second seed proves 56, 39 and 30 strict theorems, and the guided evaluation proves 58 and 42 on seeds 0 and 1 (seed 2 pending). All three seeds acquire the same families as in the first evaluation (Appendix I).

3.3 σ is rare in the knockout arm, not absent

The bases, sampled about a million times. We drew 1.1–1.5 million attempts from each knockout base, spread over the 106 strict held-out theorems (about 10^4 per theorem, with sampling on a theorem paused after its first success in each block of 1,024). Five of the six bases proved strict theorems: seed 1 seven times, seed 4 three times, seed 5 twice, and seeds 2 and 3 once each. Every success is a genuine σ proof, and twelve of the fourteen are on family members, eleven of them converse contraposition. Seed 3's single success is on the generated theorem `la_transfer_916`. Pooled rates are 0 to 6×10^{-6} per attempt; on the theorems with a success, the per-theorem estimates are 7.5×10^{-5} to 4.9×10^{-4}, each from one to three successes.

Table 5: Knockout bases at large k, pooled over the 106 strict held-out theorems (the pre-registered pool), at each seed's $K = K_{\text{train}}(16)$. Certified out of reach if $\text{UB}_{95} < 0.05/K$; $K_{\max} = 0.05/\text{UB}_{95}$ is the largest budget the data certify. In parentheses: the 105 family members only (excluding `la_transfer_916`).

knockout base	K	attempts	successes	$K_{\max}$	certified
s0	23,322	1,504,600	0 (0)	25,112 (24,892)	yes (yes)
s1	17,189	1,122,426	7 (6)	4,268 (4,730)	no (no)
s2	20,907	1,350,143	1 (1)	14,230 (14,106)	no (no)
s3	20,782	1,342,688	1 (0)	14,152 (22,376)	no (yes)
s4	21,554	1,372,403	3 (3)	8,850 (8,772)	no (no)
s5	22,297	1,432,256	2 (2)	11,375 (11,275)	no (no)

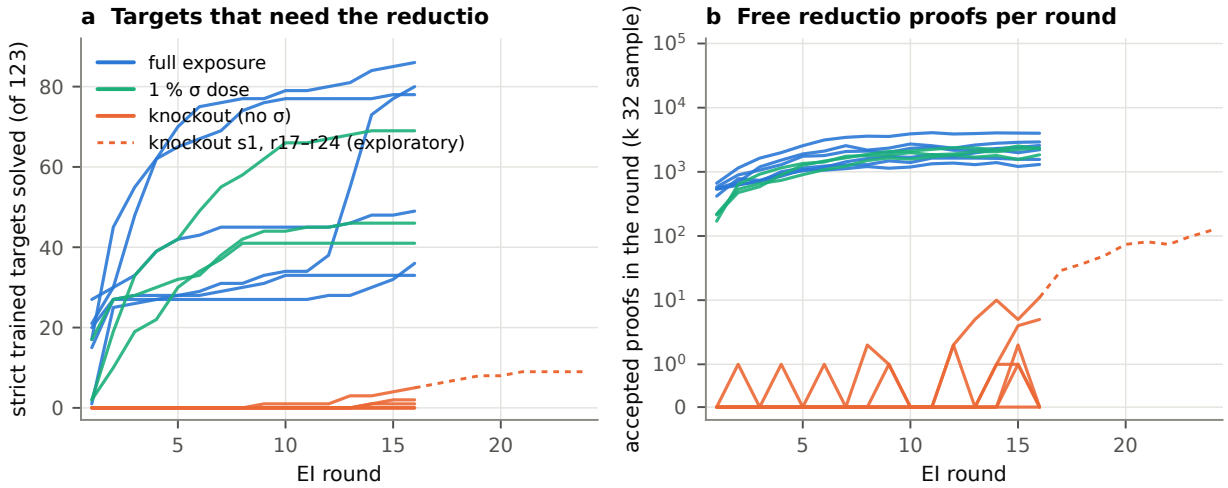


Figure 4: Ladder dynamics. (a) Strict training theorems (of 123) with an accepted proof so far. (b) Accepted proofs per round, in each round's $k = 32$ sample over all 4,495 training theorems, that use a free positive reductio (symmetric-log scale). Full and dose ladders write hundreds from round 1 and thousands later. Four knockout ladders write a few, late. Knockout s1 (dashed beyond round 16) spreads it, from 11 in round 16 to 123 in round 24.

Certificates. As pre-registered, pooled over the 106 theorems at each seed's $K_{\text{train}}(16)$, only seed 0's base is certified out of reach: 0 successes in 1.50 million attempts, $\text{UB}_{95} = 1.99 \times 10^{-6} < 0.05/K = 2.14 \times 10^{-6}$ (Table 5). Its certificate holds up to $K = 25{,}112$, 8% above the budget, and only at this one allocation and N_{eval}. Pooled over the 105 family members alone, as the experiment reported it, seed 3 is also certified; the difference is its success on `la_transfer_916`. No family is certified for any seed: at this sampling the largest certifiable K per family is 717–8,772. A pooled certificate bounds the mean rate over the pool, not any theorem in it.

The ladders. Every knockout ladder wrote σ in round 1, 6–13 distinct proofs per ladder, on theorems that supply $\neg\neg G$ or whose goal is a negation. Over their rounds, the fine-tuning mixes carried 844–4,428 such self-written σ records (2,948–9,140 σ^+). The free positive reductio was another matter. Over 16 rounds the knockout ladders produced 0, 34, 4, 1, 2 and 16 distinct proofs using it, first in rounds never, 9, 12, 15, 15 and 2. The full ladders produced 17,507–50,222 each, and the dose ladders 20,597–26,662, from round 1 (Fig. 4b). EI thus trained on σ in easy contexts from the start and did not carry the step to the theorems that need it.

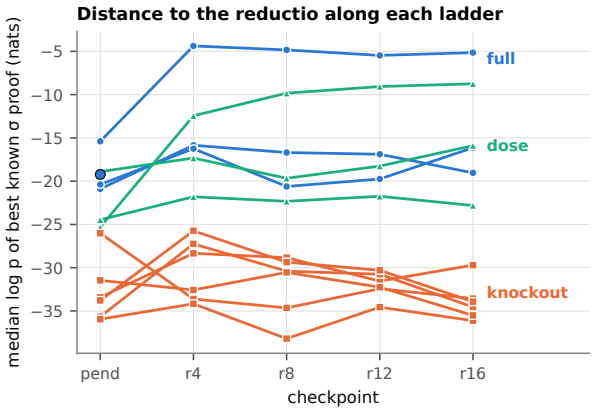


Figure 5: Median log-probability of the best known σ proof of the strict held-out theorems, along each ladder (one line-numbering offset, $T = 0.8$). The outlined point is full seed 0's base, whose ladder was not rescored.

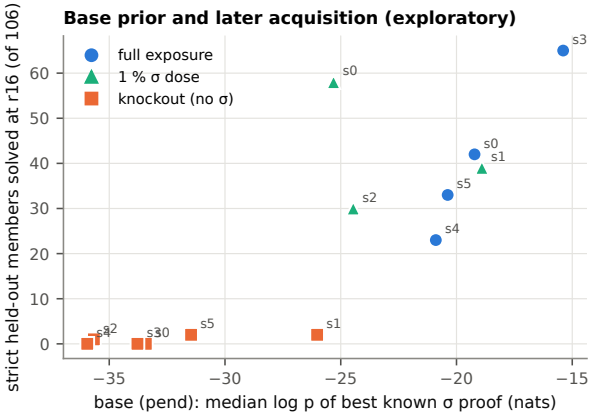


Figure 6: Exploratory (*this paper*): the base's median log-probability of known σ proofs against strict held-out theorems proved at r16, for the 13 ladders with a scored base.

Beyond the waiting-time null (*this paper*, post hoc). Suppose each knockout ladder kept sampling the 123 strict training theorems at its base's pooled rate. It would then expect a first strict success after 47 (seed 1), 115 (seed 4), 180 (seed 5) and 340 (seed 2) rounds, and never for seeds 0 and 3. The expected number of seeds with a strict success by round 16 is 0.55. Three had one: seed 1 in round 9, seeds 2 and 5 in round 14. Under the null that is a $p = 0.006$ event. So EI did raise the knockout models' rate of the free positive reductio, presumably through the σ it practised on easier theorems, but not by enough to acquire a family. The two certified-on-105 bases (s0, s3) never produced one.

Likelihood. We scored the best known σ proof of each of 101 strict held-out theorems under every checkpoint, teacher-forced at $T = 0.8$ (Fig. 5). The proofs are the full ladders' own. At r0 the median log-probability is -15.4 to -20.9 nats under the four full bases scored, -18.9 to -25.3 under the dose bases, and -26.0 to -36.0 under the knockout bases. The arms overlap only between dose and knockout seed 1. The step that costs most is the box that assumes $\neg G$: -18.0 to -21.6 nats under the knockout bases, -10.5 under full seed 0's base. From r0 to r16, two ladders move strongly toward these proofs (full s3 $+10.3$ nats, dose s0 $+16.6$). The other full and dose ladders move $+1.7$ to $+4.3$, and the knockout ladders -7.6 to $+3.7$. Several knockout ladders rise by 5–8.5 nats by r4 and fall back. Two caveats apply. The scores use one of the 33 line-numbering offsets, so they understate a proof's probability by at most $\ln 33 \approx 3.5$ nats. And a knockout ladder that found its own spelling (seed 1, below) need not raise these.

3.4 The one knockout ladder that moved

Knockout seed 1's free reductio kept spreading, so it was continued to round 24 (exploratory). Its free positive reductio proofs per round rose from 11 in round 16 to 123 in round 24. From round 21 it had proofs for 9 of the 27 strict converse-contraposition training theorems, and it never went past 9 strict training theorems in all. It proved the same five strict held-out theorems at r20 and r24, each more often: four converse-contraposition members and `la_transfer_916`. The largest gain is on `la_transfer_1019`, from 0 to 32 to 214 successes in 256 attempts at r16, r20 and r24.

No family was acquired. Its r20 proofs of these theorems take three routes: σ (32 proofs), other σ^+ (24) and the lemma route (7), a route absent from every pretraining set.

Was this elicitation of something the base already did? In the large-k sample, the base proved four of its 18 converse-contraposition members, each one to three times in 7,168–10,240 attempts. That is within reach by the rate definition, but every one is below K attempts. Three of the four members the ladder proves (`1246`, `1460`, `2256`) are among them. The fourth, `1019`, had no base success in about 10^4 attempts. A re-sample of these members at $n \geq K$ without early stopping, run for this paper, is pending.

3.5 What else the knockout changes

Data. Removing σ^+ records also thins its constituents. In K12$^-$, `DN` appears in 3.29% of records (13.73% in K12) and ¬-introduction in 5.19% (15.21%). Theorems with no intuitionistic proof fall from 4.17% to 1.35% of a 20,000-record sample; the rest use `DN` on a ¬¬ that a premise or an assumption supplies. Mean proof term size moves from 35.91 to 36.25 nodes. The knockout removes the composition's co-occurrence, and also makes each constituent three to four times rarer.

General ability. The knockout bases are 6.88 points worse at greedy decoding of 5,000 held-out generator theorems (0.8829 vs 0.9517). The gap comes from the 679 theorems whose reference proof uses σ (0.45 vs 0.96); on the other 4,321 the knockout is 0.12 points better. With 256 attempts, though, the knockout bases also prove 4.3 fewer of the 225 intuitionistic theorems of holdout250, and 4.0 fewer of textbook72's 53 (three full seeds against six knockout seeds). By r16 the holdout250 gap closes (0.996 vs 0.996), while textbook72 keeps −2.5.

Gains elsewhere. On the 322 evaluation theorems, RL's gain over what its own base reaches at RL's budget (Δ_{cov} at $K_{\text{train}}(16)$, with the base given 512 attempts and then at least K on every theorem it failed) is +37, +36 and +32 for full seeds s3–s5 and +21 to +37 for the knockout seeds: means 35.0 and 29.8, one-sided Mann–Whitney $p = 0.13$ (*this paper*). The knockout ladders are not clearly worse at RL in general, and they may be somewhat worse. These sets are not netted against RL-free controls.

What this means for the interpretation. Three things favour exposure to the composition over a general deficit. The general costs are a few theorems out of hundreds, against a gap of 8–69 vs 0–2 strict theorems gained during RL. The knockout ladders' gains elsewhere are comparable. And the dose arm restores acquisition while moving `DN` and ¬-introduction by about one point each (§3.2). The design cannot separate exposure in pretraining from exposure through replay during RL (§2).

3.6 The base's prior predicts what RL acquires (exploratory)

Across the 13 ladders with a scored base, the base's median log-probability of known σ proofs is rank-correlated with strict held-out theorems proved at r16 (Spearman $\rho = 0.88$; Fig. 6). This is mostly the difference between arms: $\rho = 0.39$ ($p = 0.38$) without the knockout. Within a ladder, the base's log-probability of a member's best known proof predicts whether the member is proved at r16. The AUC is 0.92–0.95 on the four full seeds with scored bases (pre-registered expectation ≥ 0.7) and 0.85–0.86 on the three dose seeds. Comparing only members of the same family, the AUC is 0.47–0.98 (full) and 0.28–0.60 (dose). The base's log-probability also predicts *when* a member is first proved in the ladder's per-round samples. The Spearman ρ with the first-solve round is −0.69

to −0.87 on the full seeds and −0.61 to −0.72 on the dose seeds, and −0.33 to −0.48 within families on the full seeds (all *this paper*). This agrees with reports that reference-solution likelihood predicts later success [36, 38] and that RL gains concentrate where the base already succeeds sometimes [58].

3.7 Two nulls

One seeded round. From each knockout r16 checkpoint we ran one more EI round, with and without four copies of one free positive reductio proof written by knockout seed 1's base. Seeded minus control was +0, +0, +2, +0 and +0 strict family members on seeds 0, 1, 2, 3 and 5 (seed 4 not run); seed 2's two were single successes in 256 attempts (one-sided exact $p = 0.25$). One round is too short to test seeding. In an earlier, smaller organism, a single injected proof ignited a similar pattern within one to two rounds (§6).

Power sampling. Metropolis–Hastings sampling from p^α of the full and knockout bases [21] ($\alpha = 2$, 4; 32 chains per theorem) proved none of the theorems RL r8 proves beyond the base's reach (0/33, 0/27), and none of the 106 strict theorems for knockout bases s0 and s1. Plain sampling with the same decoded tokens proved 1–2. The sampler also failed its positive control: on 17 theorems the base proves rarely, it proved 0/17, where plain sampling at the same tokens proved 6/17 ($T = 0.8$) and 13/17 ($T = 1.0$). This is a null for this sampler at these settings, not for sharpening. It agrees with the finding that practical power samplers do not reach p^α at realistic budgets [42]. Details are in Appendix F.

3.8 Robustness

The independent re-check accepted every stored knockout proof of a strict theorem in Lean 4.34.1 (448/448), the version used in the loop. It rejected all 150 re-submitted Lean rejections, all 60 proofs paired with the wrong theorem, all 40 strict knockout proofs with `Classical.byContradiction` removed, and `sorry`. Every rejection in the experiment was the pre-filter's, so the pre-filter's false-reject rate matters. Re-checking 150 rejections per reason for its 13 reasons gave 0 accepted, a 95% bound of 2.0% per reason. This paper Lean-checked every one of the 222,135 pre-filter rejections in knockout seed 1's ladder (rounds 1–16; 22,522 of them on a by-contradiction step); the result is pending. Cut-offs (a step over 512 tokens or a proof over 96 steps) can only hide successes. On the strict theorems at r16 they affect 0.04–4.97% of knockout attempts, 0.11–2.89% of full and 0.68–8.84% of dose attempts. A paired re-evaluation with doubled caps, run for this paper, moves the counts by at most one theorem so far and changes no verdict (Appendix I). The training theorems share no renaming class with any evaluation set, and no pretraining set shares one with the 106 strict theorems.

4 How much did RL create in the full-exposure models?

The knockout asks whether EI can acquire a composition the base was not shown. Most claims in the literature are instead about a single RL run and its own base. Before the knockout, the project applied the protocol of §2 to the three earlier full-exposure seeds (s0–s2, r0 vs r8, on the 322 evaluation theorems). This section summarises what it found. Every number is from reviewed reports of those runs (Appendix J).

Table 6: What survives each filter, full-exposure seeds s0/s1/s2 (r8 against its own base). The sets shrink by a factor of 3–4 from equal k to the strictest budget. The last two rows come from 9 (theorem, seed) pairs that were pre-selected (the most reliable r8 proofs in each seed's strictest set) and sampled 1.1–1.5 million times each.

filter	budget K (s0/s1/s2)	theorems
RL proves, base 0/256 (equal k)	256	54/51/60
$\mathcal{C}(K_{\text{att}})$: ladder sampling attempts on training theorems	3,574	34/33/33
$\mathcal{C}(K_{\text{train}})$: sampling + fine-tuning GPU time	7,416/7,126/9,921	33/27/24
$\mathcal{C}(K_{\text{uni}})$: all ladder GPU time, same GPU class	10,314/10,141/14,150	28/26/24
$\mathcal{C}(K_{\text{pub}})$: earlier report's value (mixed GPU classes)	19,281/21,107/23,310	22/21/14
$\mathcal{C}(K_{\text{adapt}})$: 512 each, the rest on the base's failures	44,570/43,427/56,509	21/20/14
of which no base success in $\geq$ 66,304 attempts		18/17/8
$\mathcal{C}(K_{\text{pub}})$ net of the replay-only control		20/14/8
never proved by any seed's base		4/5/5
9 pre-selected pairs: certified out of reach at K_{pub}		7 of 9 (0 of 9 at K_{adapt})
and no success from any RL-free control or other seed's base		4 pairs on 3 theorems

The budget. RL's ladder can be charged to the base in several defensible ways, and each gives a different K (Table 6). One can charge only the ladder's sampling attempts on training theorems (K_{att}), or its sampling and fine-tuning GPU time (K_{train}), or all of its GPU time (K_{uni}). One can use the value of an earlier report, which divided the A6000 ladder's seconds by a per-attempt cost measured on hard theorems on an A40 (K_{pub}). Or one can spend the budget adaptively: 512 attempts on every theorem, then the rest on the theorems the base failed (K_{adapt}). The base costs 6.36–8.04 ms per attempt on these theorems on an A40, and we convert A6000 seconds at a measured A6000/A40 ratio of 0.932. The compute-matched set is a curve in K (Fig. 7). Over a 12–16-fold range of K it falls from 34/33/33 to 21/20/14 theorems for s0/s1/s2. These 21/20/14 are compute-matched under all five allocations, and 18/17/8 of them had no base success in at least 66,304 attempts. At the level of sets, RL stays ahead of its base under every allocation and seed: Δ_{cov} is $+21/+16/+12$ at K_{pub}. Unlike the large-k crossovers reported for LLMs [56], the base here does not catch up at 10^4–5×10^4 attempts per theorem.

Brackets, certificates and controls. Sampling resolves success rates only down to about $1/n$. Below that we estimate them from likelihoods. The *known-proof estimate* sums the base's probability, teacher-forced at $T = 0.8$ over the 33 line-numbering offsets, of every Lean-accepted proof of the theorem that any model found. It recovers a median 0.98/0.94/0.93 of the measured rate on theorems with 3–100 successes in 512 attempts. It runs about two-fold low at 10^{-5}–10^{-4} (median ratio 0.45) and lower still below (0.36 on two theorems; Appendix D). Because the ratios are binned by measured rate, they condition on at least one success. Of the 9 pre-selected pairs, 7 certify at K_{pub}. Four of them, on three theorems, also had no success from the RL-free controls (512 attempts each) or from any other seed's base; only one of the three, a Peirce instance, is classical-only. The two controls receive RL's training without RL's proofs:

- *replay-only*: eight fine-tuning rounds on pretraining replay alone. It proves 239/244/249 of the 322, against the base's 232/236/234 and r8's 286/286/293.
- *continued pretraining*: about 75–78% of the ladder's GPU time. It proves 220/215/219, gaining 12–19 theorems and losing 30–42.

The cheaper control is informative: replay alone recovers 13–25% of RL's gain on the 322.

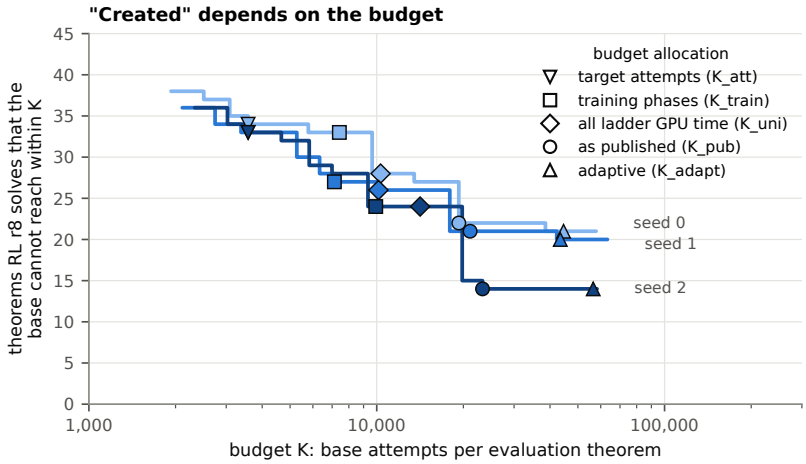


Figure 7: The compute-matched set (RL r8 proves, its base cannot reach within K) against the base budget K, for the three earlier full-exposure seeds; markers are the five allocations of Table 6. The equal-k point ($K = 256$; 54/51/60) lies off the left edge. Lines stop at 66,304 attempts, the most any theorem received; beyond it, membership cannot be decided.

Likelihoods along RL. On theorems RL newly proves, the hardest step of r8's most likely proof rises from about -6.2 nats at r0 to about -1.3 at r8, mostly by round 4 (teacher-forced at $T = 1$). Part of this is by construction, since the proof is selected at r8. RL also lifts the pretraining reference proofs of the same theorems, about three times less. On theorems neither r0 nor r8 proves, the reference proofs do not improve.

Test-time compute before and after RL. If RL mostly converts reachable-but-unlikely proofs into likely ones, extra compute at test time should help the base more than the RL model (Fig. 8). A guided evaluation adds 18–22 holdout250 theorems to the base and -1 to $+4$ to the RL models. On holdout250 the RL models are within 3–14 theorems of the ceiling, so textbook72 is the fairer pool: there, guided redraws add 8–12 to the base and 3–6 at r8. Guided attempts cost about 2.2 times the GPU time of plain ones, so this is not compute-matched. At matched sampled tokens, guided redraws still beat plain resampling on long textbook problems for six r8 models, by 5.2–10.3 points. A PUCT tree search over proof states, with a value head trained on the frozen policy's rollouts, was compared with sampling at equal wall-clock. At r8 it is within ± 3 theorems of sampling on every pool, and it failed its pre-registered gate on the theorems neither r0 nor r8 proves. At r0 it adds 10–19 theorems on 100 long theorems (13–16 lines), about 44% of what eight EI rounds add there by sampling. That fraction is 14% on holdout250 and 17% on textbook72. Neither method reaches most of the compute-matched creations: the guided base proves 3/4/1 of the 22/21/14 at K_{pub}. DeepSeek-Prover-V1.5 reported its search and RL gains as orthogonal [50]. Here they overlap: search recovers part of RL's gain at the base and adds little after RL.

5 Related work

Does RL expand what the base can do? Equal-k comparisons in LLMs find base models overtaking RL as k grows [56]. Others report gains where the base fails, read at $k = 128$ [27]. Early RL can shrink the boundary of what is solved while prolonged RL expands it [54], and wider rollouts revive plateaued runs [16]. Much of RL's measured gain corresponds to operating

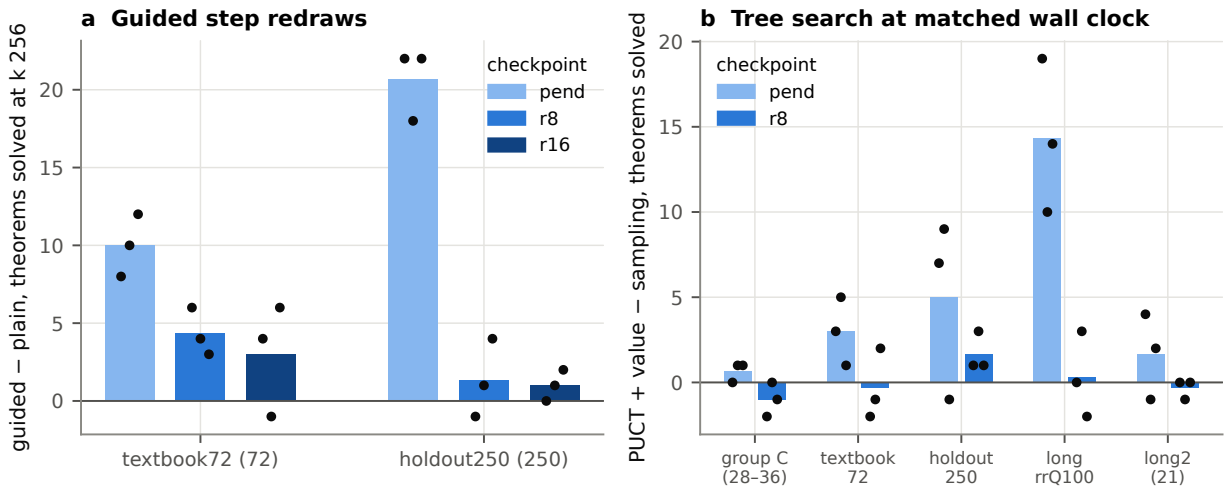


Figure 8: Test-time compute helps the base more than the RL model. (a) Guided minus plain evaluations ($k = 256$ each) for the base (r0) and after 8 and 16 rounds; bars are means, dots are seeds. Not compute-matched: guided attempts cost about 2.2 times the GPU time. (b) PUCT search with a learned value head minus sampling at equal wall-clock. "Group C" holds the theorems neither r0 nor r8 proves in an earlier evaluation (28–36 per seed); "long rrQ100" holds 100 theorems of 13–16 lines; "long2" holds 21 longer theorems.

points the base reaches with search at small sample counts [41]. Many pass@k crossovers are not statistically established [52], and reward and contamination controls change conclusions [37, 47]. In these studies the base is read at a fixed k. We tie the base's budget to RL's compute and report the created set as a curve.

Controlled organisms and pretraining exposure. The closest design is Abdulsalam et al. [1]. They pretrain on primitive rewrites that exclude later shortcuts, read the base at pass@1024 and RL at pass@16, and find GRPO building the shortcuts. Rejection-sampling fine-tuning, the family our EI belongs to, "then plateaus, while RL improves more slowly at first but continues climbing". Pretraining at a low contraction weight ρ "never reliably enters the macro or parallel regime". Their RFT and low-ρ results resemble ours; their GRPO result is the contrast. Tsilivis et al. [43] set the share of long demonstrations exactly and show RL amplifying that mode efficiently unless the share is exponentially small. Zhang et al. [58] find no RL transfer to a context with 0% or 0.1% pretraining exposure and transfer at 1%, with GRPO in a 100M model; our dose result parallels theirs for an inference rule. Pretraining with more reasoning data leaves a lead that survives RL at 8B [2]. Priming with demonstrations of a behaviour enables RL to use it [11]. RL composes new strategies only from atomic skills already mastered [7], though it composes known string functions where iterated RFT "never surpasses 2.6%" [55]. In from-scratch organisms, RL collapses onto one pretraining mode [59] and surfaces chess moves to which fine-tuning gave probability below 0.05 [38]. Small from-scratch provers and solvers also improve on their own filtered outputs [25, 30]. Our knockout keeps both constituents of the removed composition, refills the data, and measures the base's rate for the composition at about a million attempts.

Why on-policy RL may stay near the base. With an outcome reward, the number of reward queries policy gradient needs grows as the base's likelihood of a solution shrinks, and going beyond the base's support can cost exponentially much [28]. "Passive exploration" is unlikely to "generate

novel and correct proofs by chance" [49]. On-policy RL prefers solutions close to the base [39], and the KL-regularised optimum with a binary reward re-weights the base toward success [23]. On hard problems, on-policy RL rarely sees a correct rollout [34]. In GRPO on a Lean prover, gains fade at large N because rare correct proofs are seldom uplifted [14]. In LLM reasoning, EI matched or beat PPO, and RL models "fail to explore significantly beyond solutions already produced by SFT models" [13]. Our knockout measures this limit directly: the base's rate for the missing composition, and whether EI amplifies it.

Sharpening without training. Metropolis–Hastings sampling from the base's power distribution p^α approaches GRPO on MATH500 at 8.84 times the tokens [21]; a token-level approximation is reported to match one-shot GRPO [19]. Practical power samplers, however, do not converge to p^α at realistic budgets [42], and one α sharpens different problems very differently [53]. Our sampler failed its own positive control, so our null concerns the sampler.

Elicitation and measurement. A latent capability is one a model shows rarely by default and that small interventions reveal [15, 44]. Password-locked models show that a few demonstrations, or expert iteration from lucky samples, recover hidden skills, while RL from a weak policy can fail to explore [12]. Fine-tuning often enhances or wraps existing mechanisms [18, 33]. Li and Liu [26] define creation as a change in the support a model reaches under a stated budget, decoding policy and signal, and a control that never had the capability makes such claims checkable [9]. With unlimited samples, any model with full support covers everything [5]. Rare events need importance sampling or extrapolation [17, 20, 46], and reference-solution likelihoods predict success [36]. Thresholded metrics can manufacture jumps [35], so we report our family criterion beside counts. In compositional generalisation, compounds can be held out while atoms stay [22], and composing rare concepts "requires considerably more optimization steps" [29]. Our knockout holds out a compound, but it thins the atoms too (§3.5).

Expert iteration for theorem proving. Neural provers run the sample–verify–train loop with search as the expert [24, 31, 32, 48, 51] or with GRPO on a Lean reward [50]. The loop stalls when hard targets yield no successes: a from-scratch HOL prover stays at 7% without injected exploration [4], and a self-play prover found 131 correct proofs in 2.5 million attempts on unproved statements [10]. At least one formal-mathematics creation claim rests on a 64-attempt base budget [6].

6 Discussion

What the knockout shows. Without σ in pretraining or replay, 16 rounds of this EI acquired no classical family on any of six seeds. With it, full-exposure seeds gained 8–69 strict held-out theorems during RL, and a 1% dose was enough. The pre-registered verdict, NEEDS EXPOSURE, holds for converse contraposition, the family the full bases mostly had before RL; for the others it is undetermined at six seeds. Beyond the waiting-time null, the knockout ladders did raise their own rate of the free positive reductio: three reached a strict training theorem sooner than their base rates predict. They did not raise it to a family.

Four readings of the null.

1. *σ is out of the policy's reach.* This is false for five of six bases, which wrote σ proofs of strict theorems at large k. For seed 0, certified on a pooled mean rate at one budget with an 8% margin, it cannot be excluded at that budget.
2. *EI samples too little.* At 32 attempts per theorem, EI's per-round budget on the strict theorems is about 4,000 attempts, against base rates near 10^{-6}. That explains why most ladders saw nothing. It does not explain everything: three ladders reached strict theorems sooner than this null predicts ($p = 0.006$).
3. *EI amplifies rare finds too slowly, or too narrowly.* Seed 5 wrote a free positive reductio in round 2 and did not spread it in 14 more rounds. Seed 1 multiplied its free reductio more than tenfold in eight extra rounds and still proved the same five held-out theorems. One mechanism: in EI a single new proof enters a fine-tuning mix of about 70,000 RL records (up to four proofs for each of 4,495 theorems, each repeated four times) plus 20,000 replay records. Under a group-normalised policy gradient, a single success among 32 attempts would get an advantage of $\sqrt{31} \approx 5.6$ standard deviations.
4. *The knockout made the bases worse learners in general.* The general costs are a few theorems out of hundreds, and the knockout ladders' gains elsewhere (Δ_{cov} means 29.8 vs 35.0, $p = 0.13$) are not clearly smaller. The dose arm restores acquisition while changing the data much less. This reading is not excluded, but it is not needed.

Earlier organisms point the same way, and further. The project studied related patterns in smaller organisms. These are 3.2M–85M-parameter models that write whole proofs of at most six lines, trained on reductio-free or depth-limited sets. They were judged by an older in-house checker, with Lean confirmation where stated, and their reviewed results carry that label.

- **The base rate decided ignition.** Four of eleven reductio-free bases emitted the strict reductio before any RL (8×10^{-6}–2.3×10^{-4} per sample), and exactly those four ignited under EI. The other seven stayed at 0 of 606 targets. First proofs arrived about when $1/(rkN)$ predicts: rounds 1, 2, 6 and 6 against 0.2, 1.7, 2.6 and 6.2.
- **Exploration knobs did little; injected proofs did a lot.** One round at $k = 128$ or $T = 1$ changed almost nothing. One fine-tuning step on a sibling's found proofs ignited all 11 dead arms (reductio and depth-3). A single injected proof ignited the reductio on two dead seeds within one to two rounds, reaching 44 and 65 targets by round 8.
- **Size did not change it.** At 3.2M, 25M and 85M parameters, rates did not rise with size. A rare 85M draw that read 0 of 300 after eight rounds produced its first proof in round 13 when continued.
- **GRPO ignited where EI did not.** On two draws with no depth-3 proof in 600,000 base samples, GRPO acquired a depth-3 box-nesting habit in 12 of 12 arms, where paired EI did so in 1 of 3 (Lean-confirmed; the checkpoints are lost).

In the present organism, a base pretrained with no `DN` at all learned excluded middle from 16 demonstrations (35–37 of 39 members on three seeds). On the full bases, replay alone, or 16 proofs of other theorems, gave none. Taken together, in this project a pattern the base produces rarely or not at all is acquired when it is injected, even as one proof given a few rounds, or found by GRPO. EI from its own samples acquires it when the base already produces it at a rate that EI's sampling can hit. The knockout fits this account: 1% exposure put the bases above that rate, and none of six knockout bases reached it in 16 rounds. "Needs exposure" should be read as "needs exposure, or demonstrations, or a different search, for this EI".

EI or RL? The evidence on whether the null is specific to EI cuts both ways. Against: Interplay's 0% exposure null used GRPO [58], theory predicts a barrier for any outcome-reward policy gradient beyond the base's support [28, 49], and GRPO on a Lean prover rarely uplifts rare correct proofs [14]. For: RFT plateaued where GRPO built new shortcuts [1], iterated RFT failed where RL composed functions [55], and GRPO ignited from zero where EI mostly did not in our earlier organism. GRPO from the knockout bases, at matched samples, is the decisive missing experiment.

An amplification threshold? A natural hypothesis is that EI amplifies a composition once its per-theorem rate on some training theorems is high enough for 32 attempts per round to hit it within a few rounds, and acquires a family once that holds on enough members. Knockout seed 1 fits: it amplified the members its base proved at about 10^{-4}, not the family. But two doses cannot tell a threshold from a graded response. The rate in question is not the median likelihood of known proofs: dose seeds 0 and 2 (-25.3 and -24.5 nats) overlap knockout seed 1 (-26.0), and their outcomes differ by 28–56 theorems. The 0.1% dose and per-theorem rates on the training theorems would test it.

What transfers to larger models. The protocol transfers: budgets matched to RL's compute and reported as curves, RL-free controls, certificates. So does the waiting-time logic, in which a first success is expected after about $1/(rkN)$ rounds, with N the number of training problems that need the step. With 10^4–10^5 prompts and k of 8–64, LLM RL can find far rarer compositions than our 123 strict theorems allow, if a comparable share of prompts needs them. The numbers do not transfer: the 1% threshold, the 10^{-6} rates, the EI dynamics. Two practical points follow, with that caveat.

- **Evaluation.** Absence at modest k is not absence. Our knockout bases fail at $k = 256$, yet succeed at pooled rates near 10^{-6}, and on a few theorems near 10^{-4}. EI amplified the latter, on one seed, and not the former.
- **Data filtering.** Removing a composition from pretraining kept this EI from acquiring it for 16 rounds, but 1% of records restored it. We did not test the leakage levels a real filter leaves. The stronger threat, fine-tuning on demonstrations, defeated a similar filter in our earlier organism with a single proof.

An evidence hierarchy for "RL created it". In increasing strength:

1. success at equal k;
2. a compute-matched base budget, reported as a curve over defensible allocations;
3. RL-free controls with matched training;
4. a certificate against the base at that budget;
5. an intervention on what the base saw.

For our full-exposure seeds the claim shrinks at each step (Table 6). The intervention then shows that, for one composition, exposure decides whether this EI acquires it. Steps 2 and 3 cost about as much as the RL run they audit, so they are affordable wherever that run was; they are rarely reported.

7 Limitations

- **Scale and domain.** One architecture with 9.56M parameters, propositional logic over at most four atoms, and pretraining proofs of at most 12 lines. Nothing here is measured on a large

language model or on first-order logic. The organism was chosen because it allows interventions that large models do not, and its results are results about it.

- **One RL algorithm, 16 rounds, one composition.** Expert iteration at $k = 32$, $T = 0.8$ is rejection-sampling fine-tuning. GRPO-style updates, more exploration, demonstrations or longer runs could behave differently, and in the closest controlled study and in our earlier organism GRPO did (§6). One knockout ladder was still changing at round 24. Only one composition was removed.
- **What the headline test measures.** The pre-registered indicator is largely decided before RL. The RL-specific contrasts (gains during RL, families acquired during RL) were not pre-registered, and on the three full seeds that match the knockout's hardware the during-RL contrast is 2/3 vs 0/6 ($p = 0.034$).
- **Seeds and power.** Six seeds per main arm power the primary indicator but not most per-family tests (three families 3 vs 0, Holm 0.17; excluded middle 1 vs 0). A 0/6 outcome bounds the per-seed rate of acquisition from a σ-free base below 0.39 only. The dose arm has three seeds and two doses (0% and 1%) cannot locate a threshold.
- **The knockout is not surgical, and exposure includes replay.** It makes `DN` and $\neg$-introduction three to four times rarer and removes two thirds of the classical-only theorems. It costs a few theorems of general ability (§3.5). Exposure in pretraining cannot be separated from exposure through replay during RL.
- **"Absent" is not established.** Five of six knockout bases proved strict theorems with σ at large k. Only one is certified out of reach as pre-registered, on a pooled mean rate, at one allocation, with an 8% margin. No family could be certified at the sampling used, so the CREATED verdict was unreachable by design.
- **Within-reach labels.** Per-theorem base rates in the knockout arm rest on one to three successes in fewer than K attempts, from sampling that paused after a first success.
- **Estimators.** The known-proof estimate runs two- to three-fold low below 10^{-4} and was calibrated on full-exposure bases. The likelihood scores use the full ladders' proofs and one line-numbering offset.
- **Budgets.** K depends on the allocation, on the number of evaluation theorems it is spread over, and on one measured GPU-class ratio. The knockout certificates and Δ_{cov} were computed at one allocation.
- **Settings.** Old and new full seeds differ in GPU class, batch, resumption and evaluation-phase sampling, and are pooled. Many evaluations cut off more than 0.1% of attempts in some stratum (up to 8.8% of strict-theorem attempts on one dose seed); see Appendix I. The power-sampling comparators kept no proof text, so their few successes were not re-checked in Lean.
- **Interface.** In an earlier 3.2M organism, a base acting through the proof-state interface reached most of the theorems that whole-proof EI had "created", with two seeds per arm and the state confounded with step-by-step generation. Whether creation verdicts depend on the interface is untested at this scale.

8 Reproducibility

Code, data and models. All code is in the fork `dan-pandori/nd-takehome`. The experiment is on branch `dan_create-or-elicit` (commit `8e195390`, with its review): scripts `coe/` and `pod/coe/`, the independent recount code `review_coe/`, the pre-registration, per-round statistics, budgets, the estimator calibration and a claims ledger. This paper is on `dan_create-or-elicit-paper`. Its figures and statistics are regenerated by `paper/scripts/` from those files, and every number in the

text has a row in `paper/LEDGER.md` that points to its source. The pretraining sets, all 200 ladder checkpoints, the evaluation rows, the large-k samples and the Lean-gate dumps are in the public bucket `hf://buckets/dan-pandori/nd-rl/create-or-elicit/`. The decompressed md5 sums of the pretraining sets are K12 `800b5486`, K12^{-} `466fc214` and K12$^{1\%}$ `19cd2263`.

Pre-registration and review. The experiment's pre-registration was committed before its first GPU pod. Six dated amendments each precede the work they govern, except the strict-set amendment (§3.1), which came after round 1 of the first knockout ladder and before any acquisition outcome. The independent recount re-derived every count in §3 from the raw files. It corrected three Boschloo p-values that had been computed on a transposed table; no verdict changed (Appendix H). This paper's own checks and analyses were pre-registered as well (`preregistration/create-or-elicit-paper.md`, Appendix I). Two of them are post hoc, and labelled so.

Compute. Table 7 gives the compute per arm. Every 16-round ladder made 2,301,440 attempts on training theorems and 9,600 fine-tuning steps, and the knockout and dose ladders' compute is within 0.95–1.03 times the full ladders' on every quantity. The experiment used 16 GPU instances (15 NVIDIA A40 at \$0.59/h and one RTX A6000) for 309.34 instance-hours and \$182.50. This paper's checks used two A40 instances (final hours and cost: pending).

Table 7: Compute per new 16-round ladder, mean over seeds (A40; results-registry rows, recounted independently). Pretraining took a further ≈1,280 GPU-seconds, ≈24,000 steps and $\approx 4.4 \times 10^8$ tokens per seed.

quantity	KNOCKOUT	FULL (s3–s5)	DOSE
sampling GPU-s	32,441	34,005	34,492
evaluation GPU-s	7,886	8,046	7,970
fine-tuning GPU-s	13,747	13,753	13,898
total GPU-s	54,105	55,841	56,390
generated tokens	6.69×10^8	6.73×10^8	6.76×10^8
Lean checks	2.15×10^6	2.21×10^6	2.18×10^6

Acknowledgements

Left for Dan.

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A Glossary

ladder; rN; r0
one 16-round expert-iteration run from one pretrained seed; the model after N rounds; the pretrained base.

evaluation; x0, x1; guided
$k = 256$ independent attempts per theorem at $T = 0.8$; two sample seeds; the same with logically rejected steps redrawn (at most 10 per attempt).

σ, σ^+, lemma route, free positive reductio, classical-only, strict
see Table 1.

acquired
at least 50% of a family's strict held-out members proved in one evaluation.

within reach / not reached / certified at K
$\hat{p} \geq 1/K$; 0 successes in $\geq K$ attempts; one-sided 95% Clopper–Pearson bound $< 0.05/K$ (§2).

Δ_{cov}
theorems RL proves at $k = 256$ minus theorems the base reaches within K.

K12, K12$^-$, K12$^{1\%}$
the full, knockout and 1% dose pretraining sets (155,000 records each).

B Construction of the knockout set

K12$^-$ removes every K12 record whose written or pruned proof applies `DN` to a derived double negation: 16,961 records, of which 16,703 are σ and 258 cite an $\to$-elimination, $\vee$-elimination, $\bot$-elimination, `DN` or $\wedge$-elimination line. It refills 16,961 records drawn uniformly, length by length, from K12's own unused source pools: lengths up to 6 from the depth-3 generator pool and 7–12 from the long-proof pool. Every refill record passes every K12 filter (no box depth ≥ 3; `DN` only on assumptions or premises) and contains no σ^+ and no lemma route. Under an order-sensitive and a premise-order-invariant renaming key, the refill shares no class with K12 or with any of the 100 data and target files of the repository. One of K12's 11 exclusion pools was absent when the set was built and could not be checked. Every refill proof, translated to Lean, is accepted (16,961/16,961). Unchanged proofs pass the same harness and goal-negated proofs fail it. Per-length counts are identical to K12's. The dose set K12$^{1\%}$ swaps 1,550 refill records for removed σ records at K12's σ length mix. A 0.1% set is built but unused.

C When each ladder first trained on each family

First round in which at least half of a family's strict training theorems had an accepted proof in the ladder (every training theorem receives 32 attempts per round; "–" = not within 16 rounds).

ladder	conv. contrap.	neg. cond.	Peirce	Peirce seq.	excl. middle	De Morgan
FULL s0/s1/s2	1/2/1	4/13/3	–/–/3	–/–/5	–/14/–	never
FULL s3/s4/s5	2/1/1	2/–/–	8/–/–	3/–/16	never	never
KNOCKOUT s0–s5	never	never	never	never	never	never
DOSE s0/s1/s2	2/1/3	3/7/7	12/–/–	7/–/–	never	never

D Budgets at r16, the number of evaluation theorems, and the estimator

At r16 the five allocations for full seeds s0/s1/s2 are $K_{\text{att}} = 7{,}147$, $K_{\text{train}} = 17{,}531/15{,}884/20{,}422$, $K_{\text{uni}} = 24{,}619/22{,}827/29{,}510$, $K_{\text{pub}} = 47{,}962/49{,}428/50{,}447$ and $K_{\text{adapt}} = 109{,}875/100{,}929/119{,}575$ base attempts per theorem. K_{adapt} exceeds the 66,304 attempts most hard theorems received, so the r16 sets at K_{adapt} are mostly undetermined.

Every K scales as $1/N_{\text{eval}}$. For the knockout certificates (Table 5), spreading each ladder's training-phase compute over the 106 strict theorems instead of the 322 evaluation theorems would triple K (seed 0: 70,846) and void seed 0's certificate. Spreading it over 1,054 theorems would cut K to 7,125 for seed 0; at that budget seed 0 certifies easily, while the bases with successes still do not.

Figure 9 shows the known-proof estimator's calibration. Because the ratios are binned by measured $\hat{p}$, they condition on at least one success, which inflates $\hat{p}$ in the lowest bins.

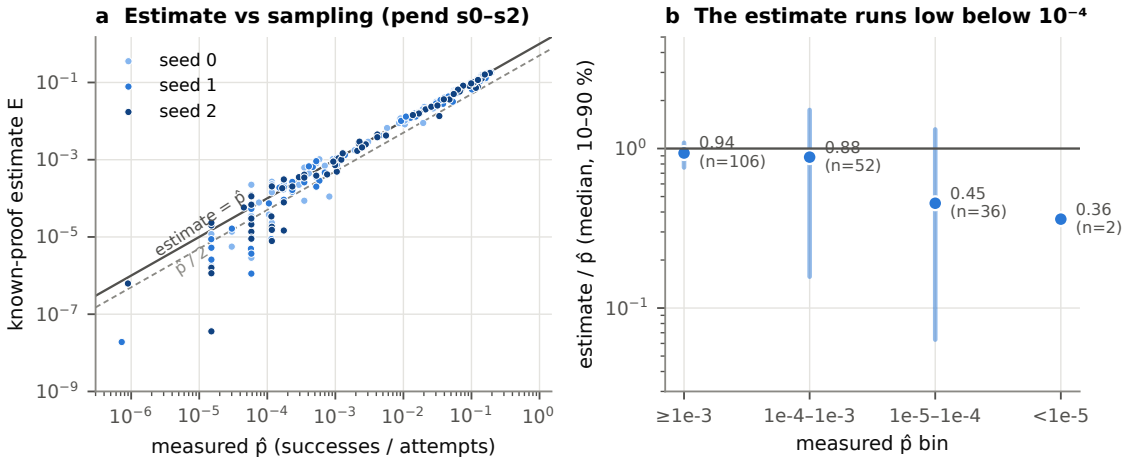


Figure 9: Known-proof estimator, full-exposure bases s0–s2. (a) Estimate against measured $\hat{p}$ for (seed, theorem) pairs with at least one success in large-k sampling. (b) Median and 10–90% range of $E/\hat{p}$ by $\hat{p}$ bin: 0.937 (106 pairs), 0.883 (52), 0.454 (36) and 0.36 (2).

Other measurements on the full seeds. At matched gains in item-response-theory ability, EI flags no more items as differentially improved than replay-only training. Eight RL rounds move the weights about as far as replay-only training (relative Frobenius norm 0.28 vs 0.30, one seed). Twelve operational definitions of "created" agree poorly: the mean pairwise Jaccard similarity of their sets is 0.29 at r8 and 0.34 at r16.

E Two more knockout-base proofs in Lean

Both are large-k successes of knockout bases before any RL. Lean accepts each and rejects it with the goal negated (`paper/scripts/lean_examples.py`). (a) Knockout seed 1's base on `la_transfer_1246`, a held-out converse-contraposition member: σ, then `DN` in both branches of an $\vee$-elimination.

```
theorem t (P Q R S : Prop) (h1 : ((¬S) → (¬((S   R)   R)))) : (((S   R)   R) → S) := by
  have n1 : ((¬S) → (¬((S   R)   R))) := h1
```

```
  have n12 : (((S   R)   R) → S) := (fun (n2 : ((S   R)   R)) => by
    have n6 : (¬(¬S)) := (fun (n3 : (¬S)) => by
      have n4 : (¬((S   R)   R)) := n1 n3
      have n5 : False := n4 n2
      exact (n5 : False))
    have n11 : S := Or.elim n2 (fun (n7 : (S   R)) => by
      have n8 : S := Classical.byContradiction (fun hh => n6 hh)
      exact n8) (fun (n9 : R) => by
      have n10 : S := Classical.byContradiction (fun hh => n6 hh)
      exact n10)
    exact n11)
  exact n12
```

(b) Knockout seed 0's base on `la_transfer_916` at $T = 1$: the refutation is built by $\to$-introduction (`n6 : ¬S → False`), not $\neg$-introduction, which Lean accepts because $\neg\neg S$ unfolds to $\neg S \to$ False. No pretraining set contains `DN` on such a line.

```
theorem t (P Q R S : Prop) (h1 : ((¬S) → (¬P))) (h2 : P) : (R → S) := by
  have n1 : ((¬S) → (¬P)) := h1
  have n2 : P := h2
  have n6 : ((¬S) → False) := (fun (n3 : (¬S)) => by
    have n4 : (¬P) := n1 n3
    have n5 : False := n4 n2
    exact n5)
  have n9 : (R → S) := (fun (n7 : R) => by
    have n8 : S := Classical.byContradiction (fun hh => n6 hh)
    exact n8)
  exact n9
```

F Power sampling

The sampler targets p^α over whole proofs by Metropolis–Hastings with suffix resampling: 32 chains per theorem, blocks of 4 proof steps, 8 MH steps per block (32 in a convergence check), $\alpha \in \{4, 2\}$ (and 1.5 in the control). Acceptance rates were 0.55–0.74. On a development pilot the per-chain success was 0.752 at 8 MH steps per block and 0.760 at 32. The plain comparators received exactly the decoded tokens of the power runs and took 1.6–2.7 times less wall-clock. They kept no proof text, so their 1–2 successes could not be re-checked in Lean.

G Pre-registered expectations against outcomes

The experiment's pre-registration stated numeric expectations. Tallied independently, they *hit* on:

- the shape of K12$^-$;
- the general-ability gap on held-out greedy decoding and on non-σ theorems;
- 0/123 strict training theorems in round 1 and 0/106 held-out strict theorems at r0 on all knockout seeds;
- knockout median log-probabilities at or below -15 nats;
- converse contraposition acquired on 3/3 new full seeds;
- at least one family on 6/6 full and 0/6 knockout seeds, the primary p and the arm medians;
- recovery of non-classical holdout250 by r16;
- $\Delta_{\text{cov}} > 0$ in both arms;
- the dose arm's round-1 σ proofs and its families;
- knockout seed 1's extension.

They *missed* on:

- the classical-only share falling by $\geq 70\%$ (observed 67.6%);
- textbook72 within ± 4 at r0 (observed -6.5 over all 72 theorems with six full seeds, two read at another batch; -4.0 on its 53 intuitionistic theorems with three);
- certificates on $\geq 5/6$ knockout bases (observed 2/6 on the 105 family members, 1/6 on the pre-registered 106);
- negated conditional on $\geq 2/3$ new full seeds (observed 1/3);
- the first σ^+ proof after round 4 (it came in round 1, on theorems supplying $\neg\neg G$);
- Δ_{cov} within +8 to +35 (two seeds at +36 and +37);
- 3–15 strict training theorems in the dose arm's round 1 (observed 2/17/2);
- a seeded-round gain on ≥ 3 seeds (observed 1/5).

The power-sampling expectations (0 solves) were met, but the sampler failed its positive control, so they are uninformative rather than hits; so is "0/106 for every sampler" (plain sampling at $T = 1$ solved one).

H Corrections applied

Three exploratory Boschloo p-values in the experiment's reports came from a transposed table. With the arms as columns, the RL-only test is $p = 0.0032$ (reported 0.0020), excluded middle 0.39 (0.30) and the three-versus-none family tests 0.034 (0.035). The primary $p = 0.00024$ and the dose test $p = 0.0033$ are unchanged. `paper/scripts/stats_extra.py` recomputes all of them. This paper also reports the knockout certificates on the pre-registered 106-theorem pool: the experiment's reports and their review used the 105 family members, which certifies seed 3 as well (Table 5). The review's ten rewordings are applied throughout. The most important replace "need" with "not acquired by this EI within 16 rounds", use "reach" only in its technical sense, and state the sharpening null for one sampler.

I This paper's checks and analyses

This writing run pre-registered its own checks (`preregistration/create-or-elicit-paper.md`, commit `c602e43c` and Addenda 1–4) before running them. The pod checks used two NVIDIA A40 instances. All evaluations are $k = 256$, $T = 0.8$, batch 512, on the 106 strict held-out theorems unless stated, with Lean in the loop.

C1: what the cut-offs hide. Paired evaluations with sample seed 0 on the same file, with standard caps (512 tokens per step, 96 steps) and doubled caps (1,024, 192). The draws differ only where a cap binds, apart from rare re-draws when the batch composition changes.

r16 checkpoint	proved, std → dbl	families acquired, std → dbl	cut-off share, std → dbl
knockout s0	0 → 0	none → none	0.04% → 0.00%
knockout s1	3 → 3	none → none	5.09% → 2.73%
knockout s2–s5	(pending)		
dose s2	28 → 29	cc, nc → cc, nc	9.30% → 8.58%
dose s0, s1; full s4	(pending)		

The pre-registered expectation that doubled caps would cut the cut-off share at least five-fold failed where cut-offs were large. Most cut attempts are non-terminating, and they stay cut at twice the budget. The counts barely move so far, so the cut-offs have not hidden successes that matter for any verdict.

C2: the dose arm on an equal footing. A second plain evaluation (sample seed 1) of dose r16 seeds 0, 1 and 2 proves 56, 39 and 30 strict theorems (first evaluation: 58, 39, 30) and acquires the same families on 3/3 seeds. The guided evaluation proves 58 and 42 on seeds 0 and 1 (seed 2 pending), again with the same families.

C3: textbook72 at the knockout's evaluation settings. The old full bases s0 and s1, evaluated at batch 512, prove 34 and 30 of textbook72's 72 problems, against 32 and 32 in their original batch-2,048 evaluation (seed 2 pending). Both are within the pre-registered ±4.

C4: knockout seed 1's members at $n \geq K$. (pending)

C5: the pre-filter, in full. (pending)

A2–A5: base prior, first-solve order and the waiting-time null. These are reported in §3.6 and §3.3, with their scripts in `paper/scripts/` (`figures.py prior`, `a3_member_prior.py`, `a4_first_solve.py`, `a5_waiting_time.py`). A2 was pre-registered after its inputs had been seen, and A5 was added after the first critique round; both are post hoc. A3 met its pre-registered expectation (AUC ≥ 0.7 on ≥ 3 of 4 full seeds: 4 of 4), as did A4 ($\rho \leq -0.5$ on 4 of 4).

J Provenance of earlier results

Run names refer to reviewed reports in the fork `dan-pandori/nd-takehome` (branch `dan_<run>`, file `review_<run>.md`) and to their summaries in the project's `experiment-summaries`.

- create-or-elicit: the knockout, dose and full s3–s5 ladders; the budgets and estimator calibration; power sampling. Review commits `dd8ee82e` (blind recount) and `8e195390` (verdict).
- capability-defs: the measurement case study on full seeds s0–s2 (budgets, certificates, controls, definitions, guided evaluations); the no-`DN` knockout taught by demonstrations (J6). Review `da11be78`.
- trajectory, rl-continue: full seeds s0–s2, rounds 1–8 and 9–16; likelihoods along RL.
- rl-from-ckpt: the replay-only control. guided-tts: guided redraws at matched tokens and the step checker. mcts-a: PUCT search with a learned value (review `5e70afbd`). grpo-best: GRPO variants from the same checkpoints.
- Earlier organisms (whole-proof token models of 3.2M–85M parameters, judged by the project's older checker with Lean confirmation where stated): the ignition study (review `review_ignition.md`), round 2 run 3 (minimum outside data), round 3 run 4a (reductio against model size) and the run4-grpo review.
- support-curves, support-state, claim-audit: the 3.2M interface result.

The claims ledger (`claims_ledger.md`, rows L1–L101) and `paper/LEDGER.md` map every inherited number to its reviewed source.